

Job Application Analysis Using Power BI: A Data Preparation and Visualization Workflow

Boumediene Allaa Eddine

Algeria

Email: allaboum123@gmail.com

Abstract

This paper presents the process of analyzing job application data using Microsoft Power BI. Starting from raw application tracking data provided in CSV format, we applied a series of cleaning and transformation steps, such as handling missing values, normalizing categorical fields, and restructuring tables for analytical purposes. The data was modeled in a star schema with one fact table (applications) and several dimension tables (candidates, vacancies, rejection reasons). We then developed DAX measures to compute total applicants, shortlisted, rejected, and accepted counts, as well as acceptance rates. Finally, interactive dashboards and visualizations were built, allowing recruitment managers to identify bottlenecks, monitor trends, and optimize hiring strategies. This workflow demonstrates how Power BI can turn raw recruitment data into actionable insights.

Keywords: Power BI, Data Preparation, Star Schema, Job Applications, Business Intelligence, Data Visualization.

1 Introduction

The recruitment process generates large amounts of data that, when properly analyzed, can provide valuable insights into applicant trends, rejection reasons, and acceptance rates. The objective of this study is to analyze job applications using Power BI by transforming raw CSV data from an Application Tracking System (ATS) into meaningful reports and dashboards. The process involves three stages: data preparation, modeling, and visualization.

2 Data Preparation

The raw dataset exported from the ATS contained inconsistencies and missing values. Several preprocessing steps were necessary to ensure clean and structured data:

- **Handling missing data:** Blank fields were replaced with the value `NA` to ensure consistency.
- **Normalizing categorical variables:** Variations such as “not reachable” and “Not Reachable” were standardized.
- **Structuring the dataset:** Columns were separated into logical entities (applicant details, vacancy details, rejection reasons).
- **Date formatting:** Application dates were transformed into a proper `datetime` format to allow time-series analysis.

3 Data Modeling

A **star schema** approach was adopted to optimize reporting and querying. The design included:

- **Fact Table: Applications** – containing applicant IDs, vacancy IDs, rejection reason IDs, status, and dates.

- **Dimension Tables:**

- **Candidates** – demographic and unique identifiers.
- **Vacancies** – job role, department, and title.
- **Rejected Reasons** – standardized rejection categories.

This schema enabled efficient use of relationships and facilitated the creation of DAX measures.

4 Measures and KPIs

Key metrics were calculated using DAX expressions, including:

- `Total Applicants = COUNTROWS(Applications)`
- `Shortlisted Count` = number of applicants marked as shortlisted
- `Rejected Count` = number of applicants marked as rejected
- `Accepted Count` = number of applicants accepted
- `Acceptance Rate = Accepted Count / Total Applicants`

These KPIs provided a comprehensive overview of the recruitment funnel.

5 Visualization and Insights

The Power BI dashboard consisted of several interactive visualizations:

- A bar chart showing total applicants, shortlisted, rejected, and accepted counts.
- A pie chart categorizing applicants by rejection reasons.
- A line chart displaying applications over time (by day).
- A clustered bar chart comparing total applicants and accepted counts by job vacancy.

Key findings:

- The majority of rejections were due to missing qualifications or incomplete documentation.
- The Business Analyst position received the highest number of applications.
- Overall acceptance rate was around 5.7%, indicating a highly competitive process.

6 Conclusion

By leveraging Power BI, raw recruitment data was transformed into a structured model that supports decision-making. The combination of data cleaning, star schema modeling, and advanced DAX measures enabled the development of insightful dashboards. These insights can help HR managers optimize recruitment strategies, identify skills gaps, and streamline applicant screening processes.